

Overhead-Aware Robust Pilot Assignment for Cell-Free Massive MIMO

Beam-Domain Resource Matching with Weighted Pilot Reuse

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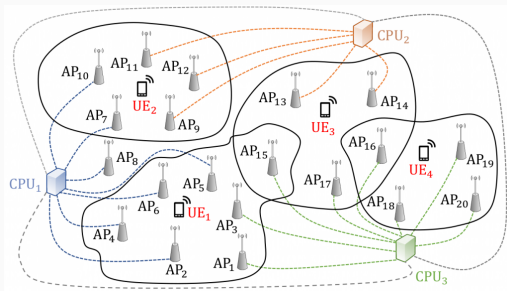
Mobile Communication Systems Term Project

System characteristic

- Many distributed APs jointly serve a smaller number of UEs ($L \gg K$), coordinated by a central processing unit (CPU) over fronthaul.
- No cell boundaries; coherent cooperative transmission yields handover-free coverage.

Why cell-free mMIMO

- Distributed APs shorten access distances and improve macro-diversity.
- Cell-free cooperation reduces cell-edge effects and SINR disparity.
- Dense AP-UE connectivity also creates many overlapping spatial links, making pilot reuse a central bottleneck.



Channel estimation via uplink pilots

A TDD coherence block of length τ_c allocates τ_p pilot symbols followed by $\tau_c - \tau_p$ data symbols. AP l receives

$$\mathbf{Y}_{p,l} = \sum_{k=1}^K \sqrt{\tau_p \rho_p} \mathbf{h}_{k,l} \phi_{p_k}^H + \mathbf{N}_{p,l},$$

where p_k is the pilot index assigned to UE k . Correlating with ϕ_{p_k} gives the observation used for MMSE channel estimation.

$$\text{SE}_k = \frac{\tau_c - \tau_p}{\tau_c} \log_2(1 + \text{SINR}_k).$$

Pilot contamination

- Typically $K > \tau_p$; UEs must reuse pilots.
- Same-pilot UEs corrupt one another's MMSE estimates.
- Contamination severity scales with the overlap of serving APs and beam directions among pilot-sharing UEs.

Design goal

Assign pilots so that same-pilot UEs share as little spatial structure as possible.

Gao (TVT 2024): AP-domain matching

- Group UEs by common serving APs.
- Many-to-many matching between UEs and APs.
- Orthogonal pilots within high-risk groups.

Limitation: AP-level matching does not resolve beam direction within a shared AP.

Mussbah (IEEE CL 2024): Beam-domain graph

- Beam-domain channel decomposition (DFT codebook).
- Active / moderate beam reports from initial access.
- Pilot assignment by graph coloring on beam overlap.

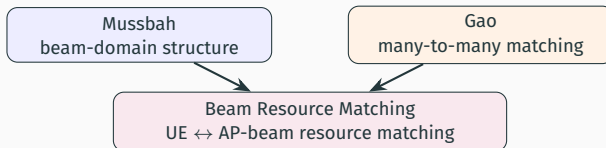
Limitation: dense beam-overlap graphs inflate the chromatic number and pilot overhead.

This work

Beam Resource Matching combines Gao's matching mechanism with Mussbah's beam-domain structure: many-to-many matching at the AP-beam resource level $r = (l, n)$.

Design principle

Beam Resource Matching lifts Gao's many-to-many matching from the AP domain to the AP-beam resource domain and inherits Mussbah's active/moderate beam structure.



- **Gao:** many-to-many matching between UEs and APs on large-scale fading.
- **Mussbah:** beam-domain conflict graph from active/moderate beam overlaps.
- **Beam Resource Matching:** matches UEs to resources $r = (l, n)$; the matching outcome determines the active beam set and feeds pilot assignment.

Beam-domain channel

$$\mathbf{h}_{k,l}^{(B)} = \mathbf{U}^H \mathbf{h}_{k,l}.$$

UE k reports beam set $\mathcal{B}_k$. The strongest beams (cumulative power threshold) form the active set:

$$\mathcal{B}_k^{(a)} \subset \mathcal{B}_k.$$

Remaining reported beams form the moderate set:

$$\mathcal{B}_k^{(i)} = \mathcal{B}_k \setminus \mathcal{B}_k^{(a)}.$$

Beam-domain conflicts capture finer interference than AP-domain.

Trade-off: dense graphs raise the chromatic number and inflate pilot overhead.

Conflict graph

$$\mathbf{B} = \mathbf{B}^{(a)T} \mathbf{B}^{(a)} + \mathbf{B}^{(a)T} \mathbf{B}^{(i)} + \mathbf{B}^{(i)T} \mathbf{B}^{(a)}.$$

$$A_{i,j} = 1 \quad \text{if } B_{i,j} > 0.$$

Pilots are assigned by graph coloring:

$$\tau_p^{\text{actual}} = \chi(G).$$

Energy efficiency

$$\text{EE} = \frac{B_w \sum_{k=1}^K \text{SE}_k}{P_{\text{tot}}},$$

$$P_{\text{tot}} = P_{\text{tx}} + P_c + N_{\text{RF}} P_{\text{RF}}.$$

Active-beam selection reduces N_{RF} through RF-chain on/off.

Core idea

Among uncolored vertices, DSATUR colors first the vertex with the largest saturation degree.

$$d_{\text{sat}}(v) = |\{c_u : u \in \mathcal{N}(v), c_u \neq -1\}|.$$

$d_{\text{sat}}(v)$ = number of distinct colors already used by colored neighbors of v .

Step	Operation	Meaning
1	Initialize $c_k = -1, \forall k$	all UEs are uncolored
2	Compute degree $d(v) = \mathcal{N}(v) $	static conflict count
3	Select uncolored vertex	choose largest $d_{\text{sat}}(v)$
4	Tie-breaking	if tied, choose largest $d(v)$
5	Assign color	smallest color unused by neighbors
6	Update saturation degree	update neighbors' d_{sat}
7	Repeat	until every UE receives a color

In Mussbah, vertices are UEs and graph edges come from active/moderate beam overlap. The number of colors becomes the actual pilot count $\tau_{\text{p}}^{\text{actual}} = \chi(G)$.

Design objective

Beam Weighted Threshold retains Mussbah's beam-domain coloring framework while suppressing weak edges through a weighted interference score.

$$B_{u,v}^w = w_{aa} |\mathcal{B}_u^{(a)} \cap \mathcal{B}_v^{(a)}| + w_{ai} |\mathcal{B}_u^{(a)} \cap \mathcal{B}_v^{(i)}| + w_{ia} |\mathcal{B}_u^{(i)} \cap \mathcal{B}_v^{(a)}|.$$

$$A_{u,v} = \begin{cases} 1, & B_{u,v}^w > \theta, u \neq v, \\ 0, & \text{otherwise.} \end{cases}$$

- w_{aa} emphasizes active-active overlap, the typically dominant interference.
- Threshold θ controls graph density.
- Pilots are assigned by graph coloring on G_θ :

$$\tau_p^{\text{actual}} = \chi(G_\theta).$$

Gao performs many-to-many matching between UEs and APs based on AP-domain large-scale fading $\beta_{m,k}$. The indicator

$$\mu(m, k) = 1$$

denotes that UE m is matched to AP k .

Quota constraints

$$|\mu(\text{UE}_m)| \leq Q_{\text{UE}}, \quad |\mu(\text{AP}_k)| \leq Q_{\text{AP}}.$$

Key setting:

$$Q_{\text{AP}} = \tau_{\text{p}}.$$

Limitation: AP-level matching cannot distinguish two UEs that share an AP through different beams.

AP group

$$\mathcal{G}_k = \{m : \mu(m, k) = 1\}.$$

Common serving AP ratio

$$M_{m_1, m_2}^{\text{AP}} = \frac{|\mu(m_1, :) \cap \mu(m_2, :)|}{|\mu(m_1, :)|}.$$

Group risk

$$S_k = \sum_{m_1 \neq m_2, m_1, m_2 \in \mathcal{G}_k} M_{m_1, m_2}^{\text{AP}}.$$

Sparse AP-domain conflict graph

$$\mathcal{T}_k(N_{\text{top}}) = \text{Top}_{N_{\text{top}}} \{\beta_{k,1}, \dots, \beta_{k,L}\}.$$

A conflict edge is placed when two UEs share at least one Top- N AP:

$$A_{i,j} = \mathbf{1}\{\mathcal{T}_i(N_{\text{top}}) \cap \mathcal{T}_j(N_{\text{top}}) \neq \emptyset\}.$$

Pilot assignment flow

- Identify the Top- N strongest APs per UE.
- Connect UEs whose Top- N sets intersect.
- Apply graph coloring on the resulting sparse graph.
- Connected UEs receive distinct pilots.

Algorithmic role

AP-Top- N provides a lightweight AP-domain sparsification of Gao's AP-overlap criterion.

Resource definition

$$r = (l, n), \quad |\mathcal{R}| = LN.$$

- l : AP index.
- n : beam index at AP l .
- Each pair (l, n) constitutes a single matching resource.

UE-resource preference

$$P_{k,r} = P_{k,l,n}.$$

Resolution gain over Gao

- Gao ranks candidates by AP-domain strength $\beta_{k,l}$.
- Beam Resource Matching ranks by beam-domain power $P_{k,l,n}$.
- Beam-level resolution separates “same AP” from “same beam.”

Modeling distinction

Matching operates at the AP-beam resource level rather than the AP level.

Reference quota

$$Q_k^{UE} = |\tilde{\mathcal{A}}_k|.$$

$\tilde{\mathcal{A}}_k$ denotes the reference active beam set from the Mussbah power-threshold rule.

Matching variable

$$\mu_{k,r} \in \{0, 1\}.$$

$$\mu_{k,r} = 1 \iff \text{UE } k \text{ is matched to resource } r.$$

Matching follows a deferred-acceptance protocol: UEs propose to their strongest resources, and each resource accepts up to Q_{res} UEs ranked by beam power.

Constraints

$$\sum_{r \in \mathcal{R}} \mu_{k,r} \leq Q_k^{UE}, \quad \sum_{k=1}^K \mu_{k,r} \leq Q_{\text{res}}.$$

$$\mu_{k,r} = 0 \quad \text{if } r \notin \mathcal{B}_k.$$

Typical choice:

$$Q_{\text{res}} = \tau_p^{\text{design}}.$$

Mussbah selects active beams via a per-UE power threshold.

Beam Resource Matching defines active beams as the matched resources:

$$\mathcal{A}_k = \{r : \mu_{k,r} = 1\}.$$

Unmatched reported beams form the moderate set:

$$\mathcal{I}_k = \mathcal{B}_k \setminus \mathcal{A}_k.$$

Algorithmic consequence

Active beams reflect global UE-resource competition, not only per-UE power ranking.

- Each beam must survive UE-resource competition.
- Resource quota Q_{res} caps UE concentration per AP-beam resource.
- RF-chain activity and pilot grouping are determined jointly by the matching outcome.

Each AP-beam resource induces a UE group from the matching outcome:

$$\mathcal{G}_r = \{k : \mu_{k,r} = 1\}.$$

Gao

AP group: $\mathcal{G}_k = \{m : \mu(m, k) = 1\}$.

Risk is computed inside each AP group.

Key replacement

Group structure shifts from AP groups (Gao) to AP-beam resource groups (Beam Resource Matching).

Beam Resource Matching

AP-beam resource group: $\mathcal{G}_r = \{k : \mu_{k,r} = 1\}$.

Risk is computed inside each resource group.

Pairwise interference between UE u and UE v is measured by a weighted overlap score:

$$W_{u,v} = w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|.$$

- $w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v|$: shared active resources; dominant interference term.
- $w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v|$: active for u , moderate for v ; secondary term.
- $w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|$: moderate for u , active for v ; secondary term.

Weight propagation

Weights propagate into the overlap ratio, resource group risk, and pilot reuse cost—not merely a diagnostic label.

Beam Resource Matching replaces Gao's common-serving-AP ratio with a beam-domain weighted counterpart:

$$M_{u,v}^B = \frac{W_{u,v}}{|\mathcal{A}_u|} = \frac{w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|}{|\mathcal{A}_u|}.$$

Unweighted baseline (Mussbah-style)

$$|\mathcal{A}_u \cap \mathcal{A}_v| + |\mathcal{A}_u \cap \mathcal{I}_v| + |\mathcal{I}_u \cap \mathcal{A}_v|.$$

All overlap types weighted equally.

Directed ratio

$$M_{u,v}^B \neq M_{v,u}^B$$

in general; denominators $|\mathcal{A}_u|$ and $|\mathcal{A}_v|$ typically differ.

Resource group risk aggregates the weighted beam-overlap ratio over UE pairs:

$$S_r = \sum_{u \neq v, u, v \in \mathcal{G}_r} M_{u,v}^B.$$

- Groups are processed in descending order of S_r .
- Within a group, unused pilots are assigned first.
- If no unused pilot remains, the pilot minimizing the weighted reuse cost is selected:

$$C(u, p) = \sum_{i: p_i = p} (M_{u,i}^B + M_{i,u}^B), \quad p_u^* = \arg \min_p C(u, p).$$

The pilot count matches the largest resource group:

$$\tau_p^{\text{BRM}} = \max_r |\mathcal{G}_r|.$$

The matching stage enforces

$$|\mathcal{G}_r| \leq Q_{\text{res}} = \tau_p^{\text{design}},$$

so the adaptive count is upper-bounded by the design budget:

$$\tau_p^{\text{BRM}} \leq \tau_p^{\text{design}}.$$

Pilot-count control

Mussbah's chromatic number grows unbounded with graph density. Beam Resource Matching caps the pilot count at the resource quota $Q_{\text{res}} = \tau_p^{\text{design}}$.

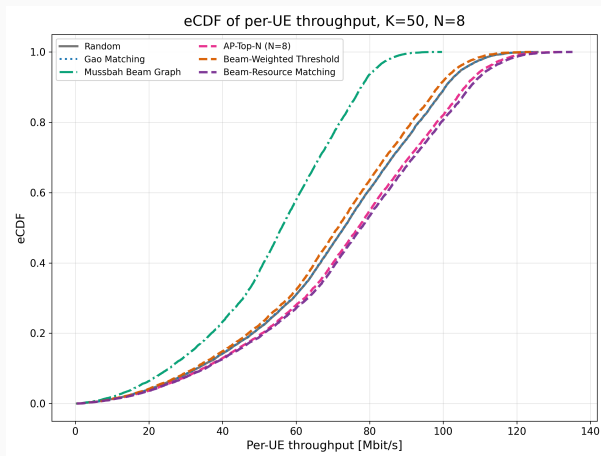
Common parameters

Parameter	Value
Number of APs	$L = 200$
Coherence block	$\tau_c = 150$
Design pilot budget	$\tau_p^{\text{design}} = 15$
Monte Carlo setups	200
Carrier frequency	3 GHz
Power control	Full power
Active-beam target	95% power
Beam-overlap weights	$(w_{aa}, w_{ai}, w_{ia}) = (2, 1, 1)$

Sweep configurations

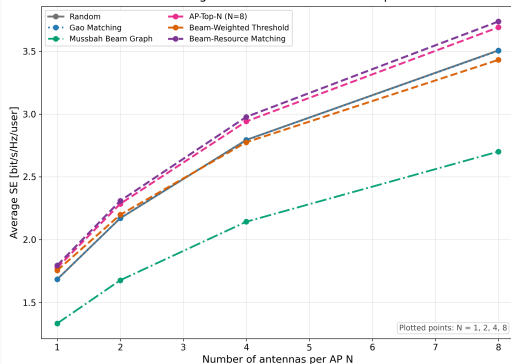
Experiment	Setting
Low- K sweep	$K = 25\text{--}50, N = 8$
N -sweep	$N = 1\text{--}8, K = 50$
High- K sweep	$K = 50\text{--}300, N = 8$
eCDF snapshot	$K = 50, N = 8$
AP-Top- N	$N_{\text{top}} = 8$
BWT edge threshold	10
Mussbah edge threshold	0
BRM resource quota	$Q_{\text{res}} = \tau_p^{\text{design}}$

Compared schemes: Random, Gao Matching, Mussbah Beam Graph, AP-Top- N , Beam Weighted Threshold, and Beam Resource Matching.



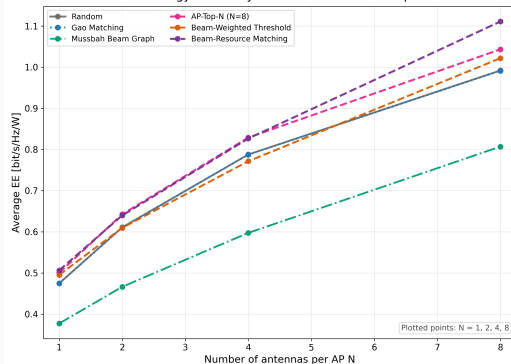
Spectral efficiency

Average SE under AP antenna sweep

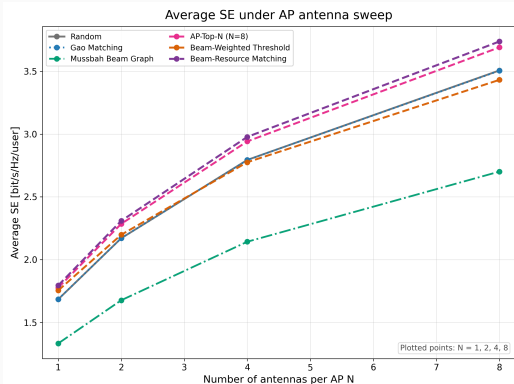


Energy efficiency

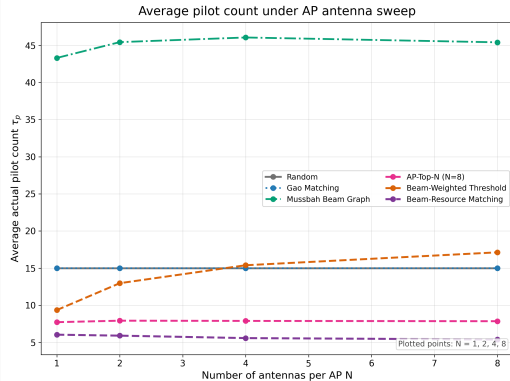
Energy efficiency under AP antenna sweep



Spectral efficiency

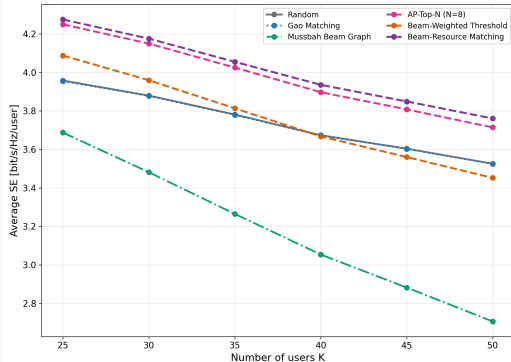


Pilot count



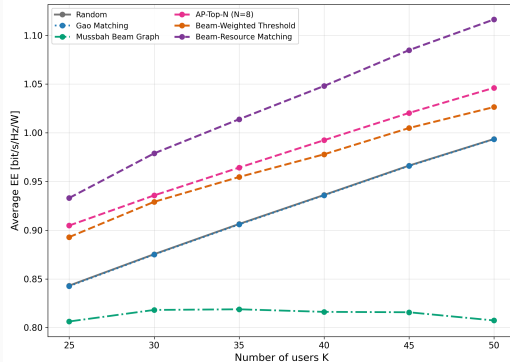
Spectral efficiency

Average SE under increasing user load



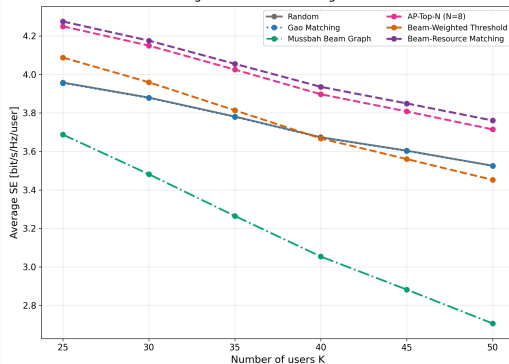
Energy efficiency

Energy efficiency under increasing user load



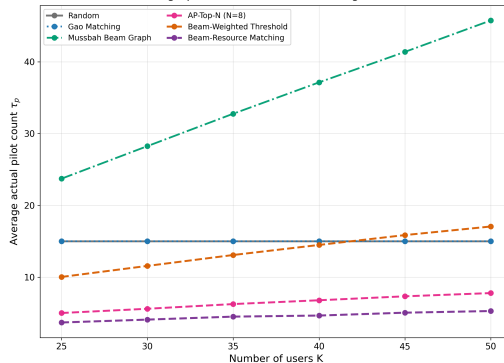
Spectral efficiency

Average SE under increasing user load

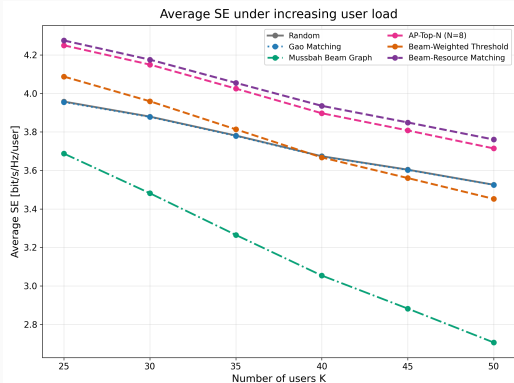


Pilot count

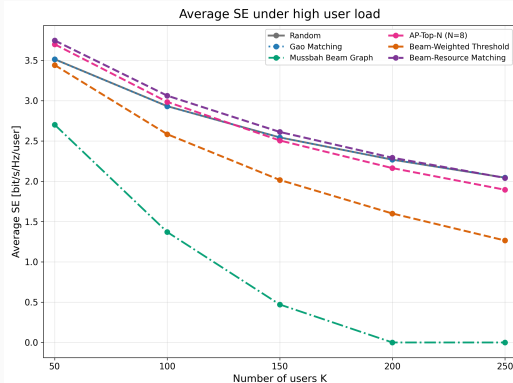
Average pilot count under increasing user load



Low-K spectral efficiency

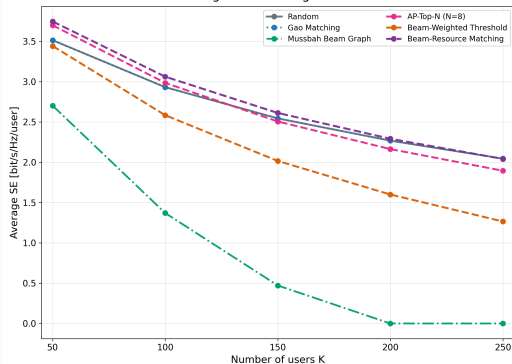


High-K spectral efficiency



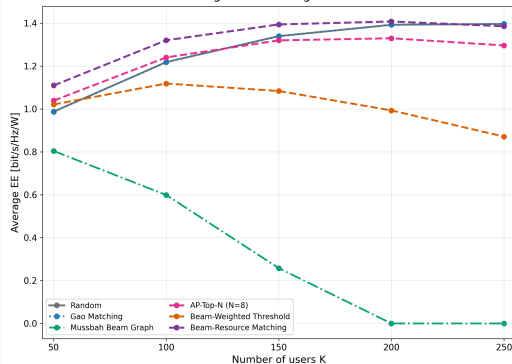
Spectral efficiency

Average SE under high user load



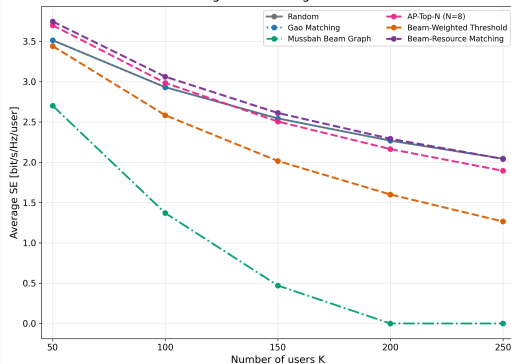
Energy efficiency

Average EE under high user load



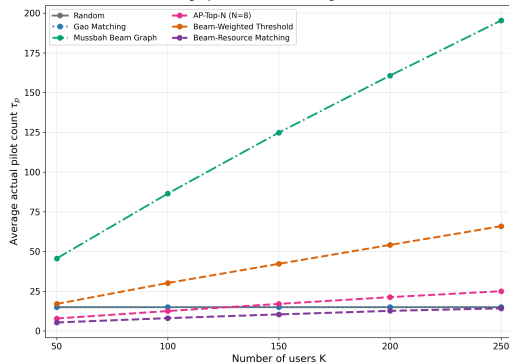
Spectral efficiency

Average SE under high user load



Pilot count

Average pilot count under high user load



1. Beam-domain is more precise

same AP $\neq$ same beam.

Two UEs can share an AP while occupying different beams.

2. Active beams are matched

- Active beams emerge from global UE-resource competition, not per-UE power alone.
- Resource quota Q_{res} bounds UE concentration per resource.

3. Conflict severity is weighted

$$M_{u,v}^B = \frac{w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|}{|\mathcal{A}_u|}.$$

Strong beam-domain collisions are prioritized during pilot assignment.

Summary

- This study proposed Beam Resource Matching (BRM) for pilot assignment in cell-free massive MIMO.
- BRM extends Gao's UE-AP matching to the AP-beam resource level, allowing users sharing the same AP but using different beams to be distinguished.
- By selecting active beams through matching and assigning pilots based on weighted beam-domain overlap, BRM reduces dominant pilot contamination while keeping pilot overhead bounded.
- Simulation results show improved spectral efficiency and energy efficiency compared with existing methods.

Future Work

- Future work can extend BRM to mobility scenarios with time-varying dominant beams.
- Another direction is joint optimization of resource matching, pilot assignment, and power control for further SE/EE improvement.

- 1 Gao, Yuan, et al. “A matching-based pilot assignment algorithm for cell-free massive MIMO networks.” *IEEE Transactions on Vehicular Technology* 73.1 (2023): 1453–1457.
- 2 Mussbah, Mariam, Stefan Schwarz, and Markus Rupp. “Beam-domain-based pilot assignment for energy efficient cell-free massive MIMO.” *IEEE Communications Letters* 28.9 (2024): 2176–2180.